

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399061
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lin; Haosheng et al.

Optical component and system for simultaneous 3D hyperspectral imaging

Abstract

A compact structure for snapshot hyperspectral imaging may include an image slicer, an integral field unit, an integral field spectrograph, and multiplexed integral field spectrograph. The image slicer includes a first section having a first plurality of mirrors, each mirror of the first plurality of mirrors having a predetermined tilt in a longitudinal direction, a second section having a second plurality of mirrors, each mirror of the second plurality of mirrors having a predetermined tilt in the longitudinal direction, and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge. The integral field unit source includes a four mirror design including an image slicer. The integral field spectrograph includes an array of spectrographs and an image slicer.

Inventors:	Lin; Haosheng (Kula, HI), Bonnet; Morgan B. (Honolulu, HI)
Applicant:	Lin; Haosheng (Kula, HI); Bonnet; Morgan B. (Honolulu, HI)
Family ID:	1000008778214
Assignee:	UNIVERSITY OF HAWAII (Honolulu, HI)
Appl. No.:	18/224594
Filed:	July 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240027265 A1	Jan. 25, 2024

Related U.S. Application Data

us-provisional-application US 63391107 20220721

Publication Classification

Int. Cl.: **G01J3/02** (20060101); **G01J3/18** (20060101); **G01J3/28** (20060101); **G02B13/22** (20060101); **G02B17/00** (20060101)

U.S. Cl.:

CPC **G01J3/021** (20130101); **G01J3/2803** (20130101); **G01J3/2823** (20130101); **G02B17/008** (20130101); G01J3/0208 (20130101); G01J3/0291 (20130101); G01J3/0294 (20130101); G01J3/18 (20130101); G01J2003/2826 (20130101); G02B13/22 (20130101)

Field of Classification Search

CPC: G01J (3/021); G01J (3/2803); G01J (3/2823); G01J (2003/2826); G01J (3/0208); G01J (3/0291); G01J (3/0294); G01J (3/18); G02B (17/008); G02B (13/22)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5420681	12/1994	Woodruff	356/326	G01J 3/2823
2008/0024871	12/2007	Achal et al.	N/A	N/A
2012/0218548	12/2011	Bodkin	N/A	N/A
2017/0268927	12/2016	Beardsley	N/A	G02B 5/1842
2018/0094977	12/2017	Taccola	N/A	G01J 3/26
2019/0155007	12/2018	Steever et al.	N/A	N/A
2020/0158568	12/2019	Kasim et al.	N/A	N/A
2020/0348176	12/2019	Faraji-Dana	N/A	G01J 3/1804
2021/0131869	12/2020	Rolland-Thompson	N/A	G01J 3/2803
2022/0113192	12/2021	Prel et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2474817	12/2011	EP	G01J 3/02

OTHER PUBLICATIONS

International Search Report and Written Opinion issued Jan. 18, 2024, in corresponding International Application No. PCT/US23/28324, 12 pages. cited by applicant
International Preliminary Report on Patentability issued Dec. 18, 2024, in corresponding International Patent Application No. PCT/US2023/028324, 8pp. cited by applicant

Primary Examiner: Chowdhury; Tarifur R

Assistant Examiner: Kidwell; Kaitlyn E

Attorney, Agent or Firm: XSENSUS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to Provisional Ser. No. 63/391,107 filed on Jul. 21, 2022, the contents of which are incorporated in its entirety.

BACKGROUND

Field

(1) The present disclosure relates to optical components and systems for simultaneous, real-time 3-dimensional (two spatial $[x,y]$ and one spectral $[\lambda]$) hyperspectral imaging of a 2-dimensional spatial field.

Description of the Related Art

(2) In many areas of business and science, cameras are used which, in addition to a spatial resolution, have a spectral resolution that often goes beyond the red, green, and blue bands that human eyes can perceive. Spectrally high-resolution imaging technology, which is referred to as “hyperspectral imaging”, has been developed for these measurements. This hyperspectral imaging allows, for example, recognition and differentiation of different chemical elements based on the spatially resolved spectrum.

(3) Early hyperspectral imaging systems based on long-slit diffraction grating (or any dispersive elements such as prisms) spectrograph used a so-called “push broom” scanning, in which one dimension is used for a spatial determination and the other dimension for a spectral determination on a two-dimensional image sensor. New approaches in hyperspectral imaging and the development of higher-resolution sensors and computer hardware have made snapshot full-frame hyperspectral systems possible.

(4) Conventional hyperspectral imagers, also known as Integral Field Spectrographs (IFS), are composed of two parts: 1) an Integral Field Unit (IFU) that reformats a two-dimensional (2D) spatial field formed by an imaging system such as a telescope or a microscope into long narrow slices or sparsely populated 2D field of light sources, and 2) a conventional grating spectrograph coupled with 2D sensor to record the spectra of all of the field points simultaneously. Three types of IFUs, namely, 1) microlens arrays, 2) coherent fiber optic arrays, and 3) machined or polished glass image slicers are commonly used for the construction of IFSs, each with their advantages and limitations. The optical systems of the spectrographs of conventional IFS are usually large due to the need to support the extended long slit or the large sparsely populated small light sources formed by the IFUs. Due to the large spectrographs, the intrinsic spectral resolution these spectrographs (limited by the illuminated size of the grating) can achieve usually far exceeds the resolution required.

SUMMARY

(5) One or more embodiments are directed to optical components and systems for snapshot hyperspectral imaging in a compact structure.

(6) One or more embodiments are directed to an image slicer for use with a multispectral light source, including a first section having a first plurality of mirrors, each mirror of the first plurality of mirrors having a predetermined tilt in a longitudinal direction, a second section having a second plurality of mirrors, each mirror of the second plurality of mirrors having a predetermined tilt in the longitudinal direction, and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge.

(7) Each of the first plurality of mirrors and the second plurality of mirrors may be plane mirrors.

(8) One or more embodiments are directed to an integral field unit for use with a multispectral light source, having a four mirror design, including an image slicer having a plurality of slicer mirrors to

receive light from the multispectral light source and output a plurality of diverging light beams, a collimator mirror that collimates each of the plurality of diverging light beams from the plurality of slicer mirrors into a plurality of collimated light beams, a plurality of reimaging mirrors to output an image of each slicer mirror onto an image sensor, and a plurality of folding mirrors that direct the plurality of collimated light beams from the collimator mirror onto the plurality of reimaging mirrors.

(9) The image slicer may include a first section having a first plurality of mirrors each having a predetermined tilt in a longitudinal direction, a second section having a second plurality of mirrors each having a predetermined tilt in the longitudinal direction, and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge.

(10) Each of the first plurality of mirrors and the second plurality of mirrors may be plane mirrors.

(11) One or more embodiments are directed to an integral field spectrograph for use with a multispectral light source, including an image slicer including a plurality of slicer mirrors, and an array of spectrographs, each spectrograph associated with a corresponding one of the plurality of slicer mirrors, wherein the array of spectrographs multiplex multispectral data onto a two-dimensional image sensor.

(12) Each spectrograph in the array of spectrographs may include a slicer mirror serving as the entrance slit of the spectrograph, a collimator mirror that collimates a diverging light beam from a corresponding slicer mirror into a collimated light beam, a micro-grating that receives the collimated light beam from the collimator mirror and diffracts light into a plurality of wavelength bands, and a reimaging mirror that directs each of the plurality of wavelength bands onto the two-dimensional image sensor.

(13) Each collimator mirror may be an off-axis parabolic collimator mirror.

(14) Each row of spectrographs in the array of spectrographs may use an integrated row of off-axis parabolic collimator mirrors.

(15) Each row of spectrographs in the array of spectrographs may use an integrated row of micro-gratings.

(16) Each row of spectrographs in the array of spectrographs may use an integrated row of reimaging mirrors.

(17) The array of integral field spectrograph may a four mirror design, including a plurality of slicer mirrors in the image slicer to receive light from the multispectral light source and output a plurality of diverging light beams, a collimator mirror that collimates each of the plurality of diverging light beams from the plurality of slicer mirrors into a plurality of collimated light beams, a plurality of folding mirrors, a plurality of reimaging mirrors to output an image of each slicer mirror onto an image sensor, wherein the plurality of folding mirrors direct the plurality of collimated light beams from the collimator mirror onto the plurality of reimaging mirrors.

(18) The image slicer may include a first section having a first plurality of mirrors, each mirror of the first plurality of mirrors having a predetermined tilt in a longitudinal direction, a second section having a second plurality of mirrors, each mirror of the second plurality of mirrors having a predetermined tilt in the longitudinal direction, and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge.

(19) One or more embodiments are directed to a multiplexed integral field spectrograph including a plurality of any of the integral field spectrographs as described above.

(20) The plurality of integral field spectrographs may receive light from a single source and may further include a field divider to divide the light from the single source to be incident onto each of the plurality of integral field spectrographs.

(21) Each of the plurality of integral field spectrographs receive light from a different source.

(22) Each of the plurality of integral field spectrographs may include an array of integral field

spectrographs and may further include a field divider to divide the light from each different light source to be incident onto each of the array of integral field spectrographs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The scope of the present disclosure is best understood from the following detailed description of exemplary embodiments when read in conjunction with the accompanying drawings.
- (2) FIG. 1 is a perspective ray trace view and FIG. 2 is a perspective schematic view of a machined image slicer integral field unit (MIST) according to an embodiment.
- (3) FIG. 3 is a perspective isometric view of an image slicer according to an embodiment.
- (4) FIG. 4 is a perspective schematic view of a machined image slicer compact spectrograph (MICS) according to an embodiment
- (5) FIG. 5 is a perspective view of the MICS according to an embodiment.
- (6) FIG. 6 is a perspective ray trace view of the MICS for the chief ray of a single beam.
- (7) FIGS. 7-9 are different perspective views of the MICS chief rays for all beams.
- (8) FIG. 10 is a side schematic view of the MICS for all beams.
- (9) FIG. 11 is a top schematic view of the MICS for all beams.
- (10) FIG. 12 is a perspective ray trace view of a configuration having four MICS sharing a common light feed according to an embodiment.
- (11) FIG. 13 is a perspective schematic view of a configuration having a plurality of MICS each with dedicated light feed according to an embodiment.

DETAILED DESCRIPTION

- (12) An integral field unit is an optical device that divides a 2D spatial field into a 2D array of image elements (pixels) or long narrow slices and using a reimaging system to reformat the spatial field into a field of sparsely populated point sources or long slits to form the input source, commonly referred to as the ‘entrance slit’ of diffraction grating spectrograph, for injection into a diffraction spectrograph for use with a multispectral light source.
- (13) As shown in FIGS. 1 and 2, a machined image slicer integral field unit (MIST) **100** includes a machined image slicer **110**, an image slicer **110** that includes a plurality of slicer mirrors **115**, a collimator mirror **120**, a plurality of fold mirrors **130**, a plurality of reimaging mirrors **140**, and an exit field stop or array of exit slits, i.e., an exit port **150**, having a corresponding plurality of images of slicer mirrors **115**, which are then output to a focal plane array **160** at a sensor. Thus, the reimaging system of MISI **100** is a four mirror design, i.e., slicer mirrors **115**, fold mirrors **130** between the collimator mirror **120** and the reimaging mirror **140**.
- (14) The plurality of slicer mirrors **115** in the image slicer **110** reflects an incoming beam I into a plurality of diverging beams B' to the collimator mirror **120**, which, in turn, collimates these diverging beams B' into collimated beams B and directs the collimated beams B onto a corresponding one of the fold mirrors **130**. Light output from each fold mirror **130** is reflected and focused by a corresponding one of the reimaging mirrors **140** to image each slicer mirror **115** at and through each exit slit **150**.
- (15) In particular, each of the micro slicer mirror **115** is reimaged to a designated position in the exit port **150** using the collimator mirror **120**, e.g., an off-axis parabolic collimator mirror, to collimates the diverging beam from the slicer mirror **115** followed by a corresponding fold mirror **130**, e.g., a micro flat fold mirror, and reimaging mirror **140**, e.g., a micro spherical mirror, to refocus onto the focal plane array. In particular, each reimaging mirror **140** may be approximately one focal length away from the intermediate pupils for each collimated beam B formed by the parabolic collimator mirror **120**, such that the exit beams are effectively telecentric.
- (16) As may be seen in FIG. 3, the image slicer **110** is divided into two sections defined by a ridge

112. The slicer mirrors **115** include a first plurality of slicer mirrors **114** and a second plurality of slicer mirrors **116** divided by the ridge **112**. The first plurality of slicer mirrors **114** have a general tilt angle to direct the beams upward, and the second plurality of slicer mirrors **116** have a general tilt angle in the opposite direction of the general tilt angle of the first plurality of slicer mirror to direct the beam downward. The division of the image slicer into two sections reduce the depth of the valleys each of the section. The valleys of a conventional image slicer, i.e., an image slicer without the ridge, would have to be very deep, rendering the image slicer impractical to manufacture. However, by including the ridge **112**, depth of the valleys may be reduced. The ridge **112** allows the image slicer **110** to direct images of the slicer mirrors **116** to different fold mirror **130**/reimaging mirror **140** configurations arranged in an array. The particular design of the image slicer **110** depends on the arrangement of these other components of the MISI **100** and uses the ridge **112** to keep the angle small, as well a focal plane size of the sensor.

(17) In a particular example, the image slicer **110** may include 56×2 slicer mirrors, e.g., each with a dimension of $0.036 \text{ mm} \times 2.664 \text{ mm}$, to divide the field into a total of 112 subfields (only 6 of which are shown for clarity). The design of the image slicer **110** depends on the downstream configuration and could include additional sections with additional ridge(s).

(18) A machined image slicer compact spectrograph (MICS) **200** according to an embodiment is illustrated in FIGS. **4** to **11**. FIGS. **4** and **6** are rotated from the actual configuration which is illustrated in FIGS. **7** to **11** for better visualization of the array. As may be seen therein, the MICS **200** uses the machined image slicer **110** design and the reimaging system of the MISI **100**, but replaces the common collimator mirror **120** with individual collimator mirrors and the fold mirrors **130** with gratings. Thus, the integral field unit of the MISI **100** is converted into a mini-spectrograph array of the MICS **200**. By incorporating the gratings directly into the integral field unit, the MICS design eliminates the need to have a large, common spectrograph behind the integral field unit, thereby greatly reducing the size of an integral field spectrograph. The following describes optical designs of MICS according to different configurations, one with a single MICS, and one with four MICS to demonstrate the flexibility and scalability of this design.

(19) The MICS **200** includes the image slicer **110** that includes the plurality of slicer mirrors **115**, a plurality of off-axis parabolic mirrors (OAPs) **220**, a plurality of micro-gratings **230**, a plurality of reimaging mirrors **140**, and a focal plane array **160**. As may be seen in FIG. **4**, the plurality of slicer mirrors **115** in the image slicer **110** reflects an incoming beam I into a plurality of diverging beams B' to each of the plurality of OAPs **220**, which, in turn, collimate these beams B' into collimated beams B and direct the collimated beams B onto a corresponding one of the gratings **130**.

(20) As may be seen in the inset of FIG. **6**, the micro-gratings **230** diffract each of the collimated beams B into a plurality of constituent light beams $\lambda_{\text{sub}.1}$ to $\lambda_{\text{sub}.n}$, e.g., three (red, green, blue) for white light, which are then separately focused as beam Bg on the focal plane array **160**. Light output from the micro-gratings **230**, here, reflective gratings, are reflected by a corresponding reimaging mirror **140** towards the focal plane array **160**. Again, the particular design of the image slicer **110** will be dictated by the arrangement of these other components of the MICS **200** as well a focal plane size of the sensor.

(21) Replacing the fold mirrors **130** with gratings **230** converts each of the 4-mirror reimaging system of MISI **100** into a mini spectrograph. Further, while the reimaging system of MISI **100** has a common parabolic collimator mirror **120**, which results in a variable reflecting angle between the incident and outgoing beam on the fold mirrors **130**, the MICS **200** uses individual off-axis parabolic mirrors **220** with the apex of the parent parabola located at the center of the corresponding slicer mirror to collimate the beam reflected by each of the slicer mirrors **115**. This design makes the collimated beams from each of the slicer mirrors **115** to propagate toward each of the corresponding micro grating **230** in parallel to maintain a constant reflection angle (or the spectrograph angle) for all the mini spectrographs. Thus, an individual MICS **200** shown in FIGS. **4** to **10** includes an array of spectrographs, one for each slicer mirror **115** of the image slicer **110**.

(22) As may be seen in the particular example shown in FIGS. 4-11, the image slicer **110** may include a 12×2 slicer mirrors, e.g., each $20 \mu\text{m} \times 0.84 \text{ mm}$ in size, forming a 4×6 array of spectrographs, as may be seen most clearly in FIG. 4. All of the mini spectrographs may have an identical grating a angle and spectrograph angle $\psi = \alpha - \beta$, where α is the incident angle of the beam of the micro-gratings **230** with respect to the grating normal and β is the exit angle of the diffracted beam with respect to the grating normal. All the micro-gratings may have the same blazing angle. In this particular example, each micro-grating may be grouped as part of a grating **235**, each grating **235** containing the 14 micro gratings for the 14 mini spectrographs in the row. Further, as may be seen therein, the individual collimator mirrors **220**, the individual gratings **230**, and the individual reimaging mirrors **140** may be each be integrated along one direction in the array, here a row direction.

(23) MICS **200** is designed to utilize modern large-format focal plane arrays (FPAs) with large multiplexing capability to obtain high quality spectral information over a 2D field simultaneously in a compact space. Given a FPA with certain physical size and pixel format, the instantaneous spatial and spectral sampling size and the hyperspectral field of view (n_x , n_y , n_λ) can be adjusted depending on the requirements of the measurements. For example, larger size optics can be used to achieve higher spectral resolution. However, this will reduce the number of mini spectrographs that can be accommodated on the sensor and the instantaneous spatial field of view coverage of the IFS. Nevertheless, the compact size of MICS allows multiple MICSs to be used in a single instrument, allowing the field of view to be easily doubled or quadrupled, as illustrated in FIG. 12.

(24) FIG. 12 shows an example system **300** including four MICSs **200a** to **200d** fed by a common source, e.g., a telescope **10**. A 2×2 field divider **320** divides the telescope focal plane into four subfields, one for each MICS. The inset in FIG. 12 illustrates details of the field divider **320**. Four 2-lens folded relay systems direct the four subfields onto respective ones of the four MICSs **200a** to **200d**. In this particular example, the telescope field is divided twice, first by the field divider **320** into four sub-fields, and each sub-field divided up by the image slicer in the respective MICSs **200a** to **200d**. Additional field dividers may be cascaded as needed and additional MICS included. The image slicer **110** of the MICS would be at the focal point of the telescope **10**, either directly or through an optical relay.

(25) Another way to multiplex to increase the hyperspectral field of view is illustrated in FIG. 13, in which a system **400** includes array of smaller telescopes **20**, a corresponding array of MICSs **220** for each telescope **20**, and a corresponding array of image sensors **30**. Each MICS **220** covers a mosaic field and the mosaic fields of all the MICS are combined together to form the image across the field. Again, the image slicer **110** of the MICS would be at the focal point of each telescope **20**, either directly or through an optical relay.

(26) Alternatively, each MICS **200** in the array may include the plurality of MICS **200a** to **200d** along with a field divider **320** for each telescope **220**, i.e., using the system **300** of FIG. 12.

(27) The present disclosure is not limited to only the above-described embodiments, which are merely exemplary. It will be appreciated by those skilled in the art that the disclosed systems and/or methods can be embodied in other specific forms without departing from the spirit of the disclosure or essential characteristics thereof. The presently disclosed embodiments are therefore considered to be illustrative and not restrictive. The disclosure is not exhaustive and should not be interpreted as limiting the claimed invention to the specific disclosed embodiments. In view of the present disclosure, one of skill in the art will understand that modifications and variations are possible in light of the above teachings or may be acquired from practicing of the disclosure.

(28) The scope of the invention is indicated by the appended claims, rather than the foregoing description.

Claims

1. An image slicer for use with a multispectral light source, the image slicer comprising: a first section having a first plurality of mirrors, each mirror of the first plurality of mirrors having a predetermined tilt in a longitudinal direction; a second section having a second plurality of mirrors, each mirror of the second plurality of mirrors having a predetermined tilt in the longitudinal direction; and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge, such that each mirror in the first plurality of mirrors is at the first angle relative to the ridge and such that each mirror in the second plurality of mirrors is at the second angle relative to the ridge.
2. The image slicer as claimed in claim 1, wherein each of the first plurality of mirrors and the second plurality of mirrors are plane mirrors.
3. An integral field unit for use with a multispectral light source, comprising: a four mirror design, including an image slicer having a plurality of slicer mirrors to receive light from the multispectral light source and output a plurality of diverging light beams, wherein the image slicer includes a first section having a first plurality of mirrors each having a predetermined tilt in a longitudinal direction; a second section having a second plurality of mirrors each having a predetermined tilt in the longitudinal direction; and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge, such that each mirror in the first plurality of mirrors is at the first angle relative to the ridge and such that each mirror in the second plurality of mirrors is at the second angle relative to the ridge; a collimator mirror that collimates each of the plurality of diverging light beams from the plurality of slicer mirrors into a plurality of collimated light beams; a plurality of reimaging mirrors to output an image of each slicer mirror onto an image sensor, and a plurality of folding mirrors that direct the plurality of collimated light beams from the collimator mirror onto the plurality of reimaging mirrors.
4. The image slicer as claimed in claim 3, wherein each of the first plurality of mirrors and the second plurality of mirrors are plane mirrors.
5. A multiplexed integral field spectrograph comprising: a plurality of compact integral field spectrographs, each compact integral field spectrograph for use with a multispectral light source, each compact integral field spectrograph imaging a different spatial field from other integral field spectrographs, each compact integral field spectrograph including an image slicer including a plurality of slicer mirrors; and an array of spectrographs, each spectrograph associated with a corresponding one of the plurality of slicer mirrors; and a two-dimensional image sensor; and the plurality of compact integral field spectrographs multiplexes multispectral data from respective two-dimensional image sensors.
6. The multiplexed integral field spectrograph as claimed in claim 5, wherein each spectrograph in the array of spectrographs of each compact integral field spectrograph includes a slicer mirror serving as an entrance slit of the spectrograph; a collimator mirror that collimates a diverging light beam from a corresponding slicer mirror into a collimated light beam; a micro-grating that receives the collimated light beam from the collimator mirror and diffracts light into a plurality of wavelength bands; and a reimaging mirror that directs each of the plurality of wavelength bands onto the two-dimensional image sensor.
7. The multiplexed integral field spectrograph as claimed in claim 6, wherein each collimator mirror is an off-axis parabolic collimator mirror.
8. The multiplexed integral field spectrograph as claimed in claim 6, wherein each row of spectrographs in the array of spectrographs of each compact integral field spectrograph uses an integrated row of off-axis parabolic collimator mirrors.
9. The multiplexed integral field spectrograph as claimed in claim 6, wherein each row of spectrographs in the array of spectrographs of each compact integral field spectrograph uses an

integrated row of micro-gratings.

10. The multiplexed integral field spectrograph as claimed in claim 6, wherein each row of miniature spectrographs in the array of spectrographs of each compact integral field spectrograph uses an integrated row of reimaging mirrors.

11. The multiplexed integral field spectrograph as claimed in claim 5, wherein each of the plurality of compact integral field spectrographs includes a four mirror design comprising: a plurality of slicer mirrors in the image slicer to receive light from the multispectral light source and output a plurality of diverging light beams; a collimator mirror that collimates each of the plurality of diverging light beams from the plurality of slicer mirrors into a plurality of collimated light beams; a plurality of folding mirrors; and a plurality of reimaging mirrors to output an image of each slicer mirror to each spectrograph of the array of spectrographs, wherein the plurality of folding mirrors direct the plurality of collimated light beams from the collimator mirror onto the plurality of reimaging mirrors.

12. The multiplexed integral field spectrograph as claimed in claim 5, wherein the image slicer of each of the plurality of the compact integral field spectrographs includes: a first section having a first plurality of mirrors, each mirror of the first plurality of mirrors having a predetermined tilt in a longitudinal direction; a second section having a second plurality of mirrors, each mirror of the second plurality of mirrors having a predetermined tilt in the longitudinal direction; and a ridge extending laterally between the first section and the second, the first section being at a first angle relative to the ridge and the second section being at a second angle opposite to the first angle relative to the ridge, such that each mirror in the first plurality of mirrors is at the first angle relative to the ridge and such that each mirror in the second plurality of mirrors is at the second angle relative to the ridge.

13. The multiplexed integral field spectrograph as claimed in claim 5, wherein the plurality of compact integral field spectrographs receive light from a single source and the multiplexed integral field spectrograph further comprising a field divider to divide the light from the single source into different spatial fields to be incident onto each of the plurality of compact integral field spectrographs.

14. The multiplexed integral field spectrograph as claimed in claim 5, wherein each of the plurality of compact integral field spectrographs receive light from a different source.

15. The multiplexed integral field spectrograph as claimed in claim 5, further comprising a field divider to divide the light from each different light source to be incident onto each compact integral field spectrograph of the plurality of compact integral field spectrographs.
